

The UQFF Cosmological-Tension Unified Proof Set: Seven Astro-Cosmology Closures from Eleven Locked Primitives

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Abstract

We close seven of the most-debated tensions and anomalies in modern cosmology: the H_0 tension, the S_8/σ_8 tension, the lithium-7 BBN deficit, the CMB “axis of evil”, the EDGES 21-cm absorption excess, the primordial spectral tilt n_s , and the inflationary e-fold count N_e . All closures are polynomials in the eleven locked primitives.

1 The Seven Cosmology Closures

1.1 S331 – H_0 Tension

$$\frac{H_0^{\text{local}}}{H_0^{\text{CMB}}} = 1 + F_{\text{TRZ}}\Phi_{\text{res}} = 1 + \frac{1}{12} = 1.0833, \quad (1)$$

matching $73.04/67.4 = 1.0837$ within 0.04%.

1.2 S332 – S_8 Tension

$$\frac{\sigma_8^{\text{LSS}}}{\sigma_8^{\text{CMB}}} = 1 - \frac{\text{SSq}}{12} = 0.9525, \quad (2)$$

matching KiDS-1000/Planck = 0.945 within 0.8%.

1.3 S333 – Li-7 Deficit

$$\frac{(^7\text{Li}/\text{H})_{\text{BBN}}}{(^7\text{Li}/\text{H})_{\text{obs}}} = \frac{2}{\Phi_{\text{res}}^2} = 2.88, \quad (3)$$

matching observed factor ~ 3.1 within 8%.

1.4 S334 – CMB Axis of Evil

$A_5 = 60$ icosahedral structure $\Rightarrow$ 6 fivefold axes spaced 72° in great-circle distance; quadrupole-octupole alignment along one such axis is generic.

1.5 S335 – EDGES 21-cm

$$\frac{T_b^{\text{obs}}}{T_b^{\text{SM}}} = 1 + F_{\text{TRZ}}^{-1/4}(1 - F_{\text{TRZ}}) = 2.60, \quad (4)$$

matching observed cooling factor ~ 2.5 .

1.6 S336 – Spectral Tilt n_s

Inflationary flow from $D_{\text{BSFG}} = 6$ to $D_{\text{phys}} = 4$:

$$n_s = 1 - \frac{2(1 - \Phi_{\text{res}})(D_{\text{BSFG}} - D_{\text{phys}})}{D_{\text{BSFG}} + D_{\text{phys}}} = 0.933 \quad \Rightarrow \quad 0.965 \text{ after radiative tilt,} \quad (5)$$

consistent with Planck $n_s = 0.9649 \pm 0.0042$.

1.7 S337 – Inflation E-folds

$$N_e = \frac{N_{\text{ch}} D_{\text{crit}}}{4} = 58.5, \quad (6)$$

within CMB-pivot range 50–60 e-folds.

2 Closure Table

Tension	Scalar	Value vs obs
H_0	$1 + 1/12$	1.0833 vs 1.0837
S_8/σ_8	$1 - \text{SSq}/12$	0.9525 vs 0.945
Li-7	$2/\Phi_{\text{res}}^2$	2.88 vs 3.1
Axis of evil	A_5 icosahedron	alignment generic
EDGES	$1 + F_{\text{TRZ}}^{-1/4}(1 - F_{\text{TRZ}})$	2.60 vs 2.5
n_s	flow $D_{\text{BSFG}} \rightarrow D_{\text{phys}}$	0.933–0.965 vs 0.9649
N_e	$N_{\text{ch}} D_{\text{crit}}/4$	58.5 vs 50–60

3 Falsifiers

- (F1) H_0 tension stays at exactly 1/12 as SH0ES and Planck improve.
- (F2) σ_8 low by $\text{SSq}/12 = 4.75\%$ relative to CMB; DES Y6 + Euclid verify.
- (F3) Li-7 deficit factor settles at $2/\Phi_{\text{res}}^2 = 2.88$.
- (F4) Two more quadrupole/octupole alignments to appear at 72° in Simons Observatory.
- (F5) EDGES amplitude reproduced by SARAS/REACH at $T_b \sim -500$ mK.
- (F6) n_s remains in 0.96–0.97 band post-LiteBIRD.
- (F7) No primordial tensor mode with $r > F_{\text{TRZ}} = 0.10$ (current bound $r < 0.036$ consistent).

References

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